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Environmental techno-economic assessment of CO_2 maritime transport and offshore sequestration

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ABSTRACT

Achieving carbon credits and enabling carbon dioxide removal (CDR) require the permanent isolation of CO_2 from the atmosphere. Ship-based CO_2 transport and injection provide flexible and scalable options for offshore storage. However, these systems are capital-intensive, requiring integrated environmental and economic assessments. This study applies life cycle assessment (LCA) and environmental techno-economic assessment (eTEA) to offshore CO_2 ship transport and injection systems in accordance with ISO/TS 14076. The analysis considers injection methods (offshore and onboard) and onboard capture. Results show that systems integrating onboard injection with onboard capture achieve the lowest global warming potential (GWP), approximately 60% lower (8.7 $kgCO_2/tCO_2$) than onboard injection systems without capture. Energy and exergy analyses further show that ship propulsion fuel consumption dominates system energy demand and thermodynamic losses. Economically, the minimum CO_2 injection price (MCIP) ranges from \$68.5/tCO_2 to \$105.6/tCO_2 depending on injection infrastructure and onboard capture. Offshore injection configurations exhibit MCIP values approximately 43–49% higher than onboard injection systems, primarily due to higher capital costs. Sensitivity and uncertainty analyses identify capital expenditures—particularly ships and offshore injection facilities—as the dominant cost drivers. These findings highlight the importance of system configuration and capital structure in designing cost-effective and environmentally sound CO_2 transport and sequestration systems.

1. Introduction

The maritime industry is the backbone of international trade, handling nearly 80% of global cargo by volume; however, its associated greenhouse gas (GHG) emissions remain a major barrier to climate change mitigation (Ahmed et al., 2023; Roussanaly et al., 2014a; Visonà et al., 2024). According to a recent report by the International Council on Clean Transportation (ICCT), tank-to-wake CO_2 emissions from global shipping reached approximately 911 Mt in 2023, with international shipping accounting for about 86% of the total (ICCT, 2025). Similarly, the World Economic Forum (WEF) Net-Zero Industry Tracker 2024 reported that emissions from the shipping sector remained relatively constant, at 0.86 Gt CO_2e in 2023 compared to 0.87 Gt CO_2e in 2019, with a 2% increase observed between 2022 and 2023 (WEF, 2024). This stagnation underscores the sector's difficulty in aligning with the International Maritime Organization (IMO) strategy, which targets a 70% reduction in carbon intensity by 2050 relative to 2008 levels (IMO, 2023).

To date, a range of decarbonization strategies have been proposed, including alternative marine fuels, operational measures, and vessel efficiency improvements (Robalo-Cabrera et al., 2025; Ma et al., 2025). While these approaches offer incremental emission reductions, their scalability and long-term effectiveness remain uncertain (Jimenez et al., 2022; Sou et al., 2022). Given these limitations, a complementary solution capable of delivering deeper and more immediate reductions is required. Carbon capture and storage (CCS) has therefore garnered

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attention as a viable measure for achieving large-scale, near-term emission reductions by capturing CO_2 and transporting it to secure storage sites (IEA, 2023; IPCC, 2022; Zhang et al., 2023). Although current CCS projects are primarily focused on large stationary sources—such as power generation, cement, and steel production facilities (DNV, 2025; McKinsey and Company, 2025)—transport infrastructure remains a critical bottleneck in enabling full-chain CCS deployment (Vakili et al., 2023; Vakili et al., 2025).

Compared with alternative fuels, operational measures, and energy-efficiency technologies, onboard CO_2 capture offers a distinct decarbonization pathway by directly reducing combustion-related emissions from existing fuel-based ship operations. Although alternative fuels are essential for long-term maritime decarbonization, their near-term deployment is constrained by fuel availability, bunkering infrastructure, cost, and vessel compatibility. In this context, onboard CO_2 capture is relevant as a transitional option, particularly when integrated with ship-based CO_2 transport and offshore storage infrastructure.

CO_2 can be transported via pipelines or ships, each presenting distinct trade-offs (Holm et al., 2025). Pipelines are well suited for continuous, large-volume flows but face long lead times, complex permitting processes, and limited flexibility for cross-border applications (Rui et al., 2025). In contrast, CO_2 shipping offers modularity, regional adaptability, and cost advantages, particularly for early-stage CCS networks and long-distance transport (IEAGHG, 2020; Element Energy, 2018). As CCS deployment expands geographically, the need for flexible transport solutions becomes increasingly urgent. Projections suggest that CO_2 shipping demand in the Asia-Pacific region alone could exceed 100 Mt annually by 2050, necessitating over 100 dedicated carriers (S&P Global, 2024; Global CCS Institute, 2022). These forecasts position CO_2 shipping not as a supplementary option, but as a central enabler of future CCS infrastructure.

Despite the growing interest in ship-based CCS, academic and industrial studies on CO_2 shipping remain fragmented. Most prior research has focused on techno-economic analyses (TEA), primarily benchmarking costs for pipeline and ship-based transport chains (Knoope et al., 2013; Roussanaly et al., 2014b; Deng et al., 2019; d'Amore et al., 2024; Vakili et al., 2025). In contrast, studies dedicated solely to environmental life cycle assessment (LCA) of CO_2 transport systems are limited, and, to the best of our knowledge, no LCA has been conducted on ship-based systems incorporating onboard CO_2 capture. Moreover, no study has presented an integrated LCA–TEA for ship-based CCS, leaving the environmental and economic performance of these emerging configurations largely unexamined. Research addressing innovative options such as offshore injection facilities or onboard capture is similarly scarce. As ship-based CCS must achieve both environmental integrity and economic feasibility for large-scale deployment, a unified assessment framework is essential to identify design configurations that minimize trade-offs and avoid burden shifting (Zhao et al., 2025).

This study addresses these gaps by integrating LCA and environmental techno-economic assessment (eTEA) to evaluate large-scale, long-distance CO_2 ship transport and offshore injection systems. The analysis explicitly incorporates both conventional offshore injection facilities and advanced onboard injection and capture options, in accordance with ISO/TS 14076 (ISO, 2025), ensuring methodological consistency and comparability. This study contributes by: (1) quantifying the environmental performance of ship-based CCS systems under varying injection and capture configurations; (2) evaluating economic performance within an eTEA framework informed by LCA results; and (3) analyzing the impact of capital and operating cost support mechanisms on overall system viability. The integrated LCA–eTEA framework presented herein offers a transparent foundation for project developers and policymakers to design cost-effective and environmentally robust CCS shipping strategies.

2. Technology overview

This section provides an overview of key technologies relevant to the maritime transport and storage of CO_2, followed by a description of the system boundaries and assumptions adopted in this study. The carbon capture, utilization, and storage (CCUS) value chain typically comprises: (1) CO_2 capture, (2) compression and liquefaction, (3) transport, and (4) storage or utilization. This study focuses on the post-liquefaction stages of ship-based CO_2 transport and injection, with optional consideration of onboard capture where applicable.

Liquefied CO_2 is commonly produced by cooling and pressurizing captured CO_2 to approximately –50 °C and 6–7 bar, reducing its volume by a factor of about 600 compared to its gaseous phase (Noh and Kang, 2025; Wang et al., 2025; Zahid et al., 2014). The analysis begins after liquefaction, and all scenarios are defined based on the transport and injection of liquefied CO_2.

2.1. CO_2 ship transport

Ship-based transport is regarded as the most practical option for long-distance, large-scale CO_2 shipment. Medium to large liquefied CO_2 carriers are currently under development in countries such as Japan and Norway (Mitsubishi Heavy Industries, 2024; TGE Marine, 2023). Typical onboard storage conditions are maintained at approximately –50 °C and 6 bar (CLIMIT, 2022; Deng et al., 2019), with vessel capacities ranging from several thousand to tens of thousands of tonnes (Maersk Broker, 2022). In this study, two 20 kt-class carriers are assumed, targeting an annual CO_2 injection volume of 0.8 $MtCO_2$. The transport route spans approximately 6,300 km, from the west coast of Korea to an offshore injection site in Indonesia. The ships are assumed to operate on liquefied natural gas (LNG) at a cruising speed of 15 kn. LNG was selected as the marine fuel due to its extensive prior evaluation in the literature, offering a consistent and reliable basis for comparison (Al Ghafri et al., 2021; Awoyomi et al., 2019).

2.2. Boil-off gas (BOG) reliquefaction

During long-distance transport of liquefied CO_2, a portion of the cargo inevitably evaporates due to heat ingress, generating BOG (Lee et al., 2017). According to Yoo et al. (2013), BOG generation typically occurs at approximately 0.15%/d, indicating that evaporation is an inherent aspect of CO_2 shipping operations. To minimize cargo loss and maintain stable pressure within storage tanks, CO_2 carriers are equipped with reliquefaction systems that capture the evaporated CO_2, compress and cool it, and return it to the cargo tanks in liquid form. In this study, CO_2 reliquefaction is treated as an integral component of the ship transport system, and its energy consumption and associated costs are included in the techno-economic assessment (Noh and Kang, 2025).

2.3. CO_2 injection

In offshore applications, CO_2 injection into geological storage formations requires high-pressure pumping systems, with two primary approaches: deployment of Floating Storage and Injection (FSI) facilities or installation of injection equipment directly aboard the ship (DNV, 2021). The FSI approach is more suitable for large-scale, continuous operations, while ship-based injection systems offer greater flexibility for adapting to varying reservoir conditions (IEAGHG, 2020). However, ship-based systems require additional space and energy, and they differ significantly from FSI systems in terms of capital and operating costs, as well as emissions (CATO, 2016). Several demonstration projects are currently underway for both methods, with ongoing efforts focused on design optimization and economic evaluation. This study incorporates both injection modes in its analysis.

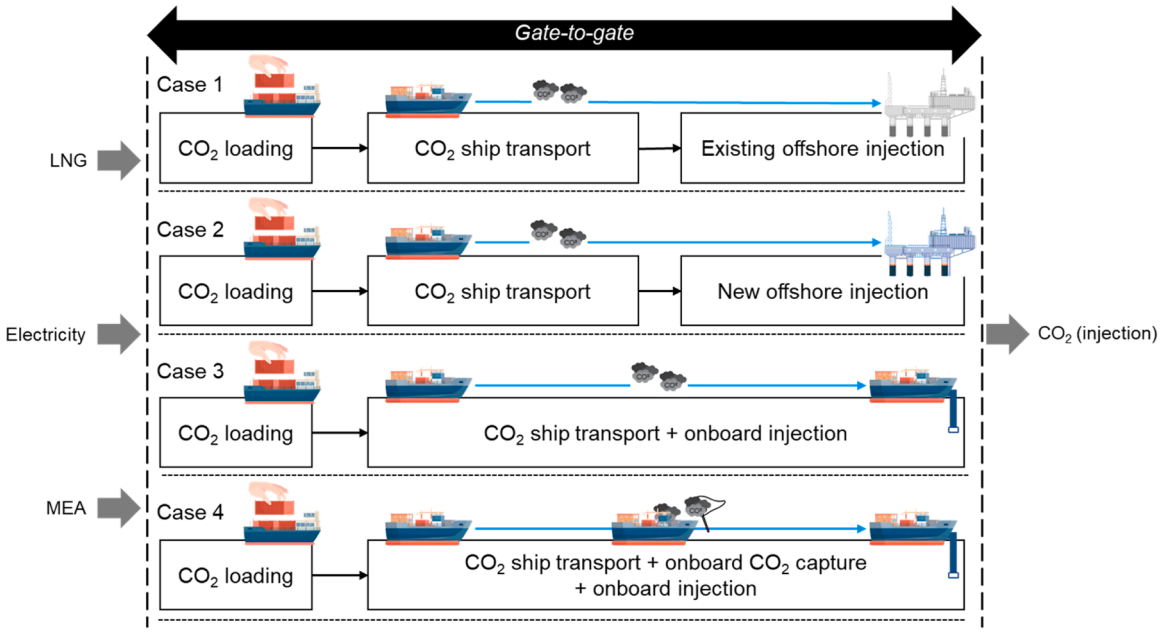


Fig. 1. System boundary for CO_2 transport and injection stages considered in this study.

2.4. CO_2 subsea storage tank

In offshore injection systems, a subsea storage tank is employed as a buffer to temporarily hold liquefied CO_2 prior to injection into geological formations. This configuration is particularly relevant when injection is conducted via offshore platforms rather than directly from ships. The subsea tank stabilizes the flow of CO_2 by accommodating fluctuations in the injection rate caused by variations in reservoir pressure or operational interruptions. In this study, the tank's storage capacity is assumed to be 1.5 times that of a single carrier, equivalent to 30,000 m^3 for the 20 K CO_2 carriers considered—ensuring it can receive a full vessel's cargo while maintaining sufficient buffer volume to prevent operational delays and support continuous injection, even when unloading schedules are not perfectly synchronized (Luo et al., 2023).

2.5. Onboard CO_2 capture

Onboard CO_2 capture technologies are generally classified into pre-combustion, post-combustion, and oxy-fuel processes, of which only post-combustion is applicable to ships (Hua et al., 2023; Gür, 2022). In this study, the representative post-combustion method—MEA-based absorption—was considered. This technology, which is already commercialized in onshore power plants (Leeson et al., 2017; Han et al., 2023), was applied to capture CO_2 from exhaust gases during ship operation, with the captured CO_2 included in the total amount transported and injected.

3. Methods

3.1. LCA

An LCA was conducted in accordance with ISO 14040 and 14044 standards (ISO, 2006a; ISO, 2006b) to evaluate the environmental impacts across the entire product life cycle. The assessment followed four standardized steps: (1) goal and scope definition, (2) life cycle inventory analysis, (3) life cycle impact assessment, and (4) life cycle interpretation.

3.1.1. Goal and scope definition

The primary objective of this study was to quantify the global warming potential (GWP) associated with the transport of liquefied CO_2 by ship and its subsequent offshore injection. The functional unit was defined as the injection of 1 ton of CO_2, serving as the reference for comparing environmental impacts across the different cases. GWP was selected as the primary environmental indicator because the main differences among the cases are associated with direct CO_2 emissions from ship operation and their reduction through onboard CO_2 capture.

3.1.2. System boundary

To focus on the environmental impacts of the transport phase, a gate-to-gate system boundary was adopted, encompassing the process from CO_2 loading to subsea injection. Upstream processes, such as CO_2 capture at industrial sources, and downstream operations related to long-term geological storage were excluded. Capital goods and infrastructure construction, including shipbuilding, offshore facility construction, onboard injection equipment manufacturing, and onboard capture unit manufacturing, were also excluded from the LCA boundary. Accordingly, the LCA results represent operational environmental impacts within the defined transport–injection boundary, rather than full infrastructure life-cycle impacts. Four cases were established based on the configuration of injection facilities and the inclusion of onboard CO_2 capture. The system boundaries for these cases are illustrated in Fig. 1.

3.1.2.1. Case 1: Offshore injection with existing facility. The first case assumes the use of an existing offshore injection facility for CO_2 transport and injection. While the environmental impacts associated with the injection facility are accounted for, the capital cost of constructing new infrastructure is avoided.

3.1.2.2. Case 2: Offshore injection with new facility. The second case involves the installation of a new offshore injection facility. As in Case 1, the environmental impacts of the injection process are included; however, this scenario requires additional capital investment. Transport of CO_2 by ship followed by offshore injection is currently considered the most practical and commercially mature model, as demonstrated by projects such as Northern Lights in Norway (Northern Lights, 2025).

3.1.2.3. Case 3: Onboard injection. The third case assumes the CO_2 injection facility is integrated onboard the ship, enabling the vessel to perform both transport and injection operations. In this configuration, compression and pumping modules are installed on the CO_2 carrier,

allowing the ship to inject liquefied CO_2 directly into offshore geological storage without using a separate offshore injection facility. Although still in the demonstration phase, this configuration is viewed as a promising approach for future market expansion and cost reduction.

3.1.2.4. Case 4: Onboard capture and injection. The fourth case builds upon Case 3 by incorporating onboard CO_2 capture, resulting in a multifunctional ship capable of capturing, transporting, and injecting CO_2. In this configuration, CO_2 emissions produced during ship operation are captured and stored, thereby reducing the overall environmental burden.

To ensure consistent comparison across the four cases, the main engineering assumptions for the CO_2 transport and injection system were defined before developing the life cycle inventory and techno-economic model. The key design basis is summarized in Table 1. The table includes the main assumptions used to calculate voyage frequency, including annual CO_2 transport volume, carrier capacity, number of carriers, transport distance, round-trip duration, loading and unloading time, and sailing speed. The CO_2 carrier capacity was defined on a volumetric basis, and the effective loading amount was calculated using the assumed liquid CO_2 density. For Case 4, the annual CO_2 transport volume includes the additional CO_2 captured onboard during ship operation.

3.1.3. Life cycle inventory analysis

The material and energy inflows, as well as the corresponding outputs and emissions within the system boundary, were quantified based on the defined functional unit and are summarized in Table 2. The life cycle inventory was developed using the ecoinvent V3.11 database, with detailed data provided in Supplementary Material Table S1. Due to the absence of Indonesia-specific datasets in the LCA database, global ecoinvent datasets were consistently used for the inventory inputs. Fuel consumption values were cross-validated against previous CO_2 shipping studies (Deng et al., 2019; Vakili et al., 2025), whereas the electricity demand for offshore injection in Cases 1 and 2 was adopted from IEAGHG (2020). For Case 4, MEA consumption was included as the make-up requirement for onboard CO_2 capture, reflecting solvent replacement during operation.

3.1.4. Life cycle impact assessment

The life cycle impact assessment was conducted using SimaPro V10.2, applying the ReCiPe 2016 (H) midpoint method (Park et al., 2025a; Valasara et al., 2025a; Valasara et al., 2026a). Among the 18 impact categories, GWP (kg CO_2) was selected as the key indicator (Valasara et al., 2026b; Park et al., 2025b; Yun et al., 2025). As the primary objective of this study is to quantify how environmental impacts translate into economic outcomes, the analysis focuses on GWP due to its direct relevance to environmental regulations and climate change mitigation efforts. To provide a broader environmental perspective, the normalized results for all 18 midpoint impact categories are provided in Supplementary Material Figure S1.

Table 1
Key engineering assumptions for CO_2 ship transport and offshore injection systems.

Parameter	Unit	Value	Applied case
Annual CO_2 transport volume*	$MtCO_2$/year	0.79	Case 1–3
		0.81	Case 4
CO_2 carrier tank capacity	tCO_2/voyage	20,000	All cases
Liquid CO_2 density	tCO_2/m^3	1.1	All cases
Number of CO_2 carriers	vessels	2	All cases
One-way transport distance	km	6,300	All cases
Round-trip duration**	h/voyage	488	All cases
Loading time	h/voyage	24	All cases
Unloading time	h/voyage	24	All cases
Sailing speed	knots	15	All cases
Voyages per year***	voyages/year	18	All cases
Project lifetime	year	30	All cases

* The annual CO_2 transport volume in Case 4 is slightly higher because CO_2 captured onboard is included in the total injected amount.

** The round-trip duration includes loading, outbound sailing, unloading, and return sailing.

*** The number of voyages per year was calculated using the annual CO_2 transport volume, effective CO_2 loading amount per carrier, and number of carriers.

Table 2
Life cycle inventory for CO_2 transport and injection normalized to 1 tCO_2 injected.

	Material	Unit	Value per 1 tCO_2 injected
Case 1	LNG fuel	kg/tCO_2	7.380
	Electricity	kWh/tCO_2	2.300
Case 2	LNG fuel	kg/tCO_2	7.380
	Electricity	kWh/tCO_2	2.300
Case 3	LNG fuel	kg/tCO_2	7.380
	Electricity	kWh/tCO_2	5.170
Case 4	LNG fuel	kg/tCO_2	7.250
	Electricity	kWh/tCO_2	5.077
	MEA	kg/tCO_2	0.029

3.1.5. Sensitivity analysis for LCA

To identify key variables influencing the LCA results and enhance the robustness of the findings, a sensitivity analysis was conducted with respect to transport distance and fuel type. Transport distance was selected due to its direct effect on fuel consumption and, consequently, GHG emissions. Fuel type was included to account for variations in emission characteristics during combustion.

3.2. Energy and exergy analysis

An energy and exergy analysis was conducted to evaluate the thermodynamic performance of the four CO_2 transport and injection configurations from a system perspective. A system level approach was adopted because detailed process simulation for individual unit operations was not performed. Therefore, the analysis focuses on the overall energy input and exergy distribution within the defined system boundary.

The energy analysis quantifies the total energy required to transport and inject 1 t of CO_2 for each configuration. The primary energy source in the system is LNG fuel used for ship propulsion, while electricity consumption is associated with onboard injection and CO_2 capture systems where applicable. The lower heating value (LHV) of LNG used in this study was assumed to be 48.6 MJ/kg (Al-Breiki and Bicer, 2020).

Exergy analysis was conducted to evaluate the quality and utilization efficiency of the supplied energy. In general, total exergy consists of physical, chemical, kinetic, and potential exergy components (Oh et al., 2025). In this study, only physical exergy and chemical exergy were considered because kinetic and potential exergy contributions are negligible relative to the dominant energy flows in the system (Kim et al., 2026).

The physical exergy is defined as

$$ex_{ph} = (h - h_0) - T_0(s - s_0) \quad (1)$$

where h and s denote the specific enthalpy and entropy, respectively, while h_0 and s_0 correspond to the thermodynamic properties at the environmental reference state. T_0 represents the ambient temperature. The reference environment was taken as 25 °C and 1 bar. The chemical exergy of LNG was estimated based on its LHV (Szargut et al., 1986).

The overall exergy balance of the system is expressed as

$$Ex_{in} = Ex_{useful} + Ex_{loss} + Ex_{destroyed} \quad (2)$$

where Ex_{in} is the total exergy input to the system, Ex_{useful} represents the useful exergy associated with the transported CO_2 stream, Ex_{loss} denotes

exergy discharged to the environment, and $Ex_{destroyed}$ corresponds to exergy destruction due to thermodynamic irreversibilities. In this study, the heat and work generated from LNG combustion for ship propulsion were included in the exergy destruction term in the system-level exergy balance.

The exergy efficiency of the system was calculated as

$$\eta_{ex} = \frac{Ex_{useful}}{Ex_{in}} \tag{3}$$

which represents the fraction of supplied exergy effectively utilized for CO_2 transport and injection.

3.3. eTEA

eTEA is an analytical framework that integrates techno-economic and environmental indicators to support informed decision-making. In this study, the eTEA was conducted in accordance with ISO/TS 14076:2025 guidelines (ISO, 2025). The economic analysis followed conventional TEA methodologies while maintaining consistency with the LCA by applying the same functional unit and system boundaries (Gong et al., 2026a; Park et al., 2024).

3.3.1. Techno-economic analysis

In this study, capital costs were estimated using data collected from official reports and various literature sources, then adjusted using the exponential scaling equation (Eq. (4)) to account for economies of scale (Ahn et al., 2026a; Yang et al., 2024), where n represents the scaling exponent that characterizes how equipment cost varies with size:

$$New\ Cost = (Base\ Cost) \times \left(\frac{New\ Size}{Base\ Size}\right)^{n} \tag{4}$$

A project lifetime of 30 years was assumed (Gong et al., 2026b; Baek et al., 2026). All capital and operating costs were converted to 2023 values using the Chemical Engineering Plant Cost Index (CEPCI) and Eq. (5) (Ahn et al., 2025a; Ahn et al., 2025b; Park et al., 2026).

$$2023\ \ Cost = (Base\ Cost) \times \left(\frac{2023\ \ Cost\ Index}{Base\ Year\ Index}\right) \tag{5}$$

The key economic indicator was the minimum CO_2 injection price (MCIP), defined as the break-even price required to achieve the target internal rate of return (IRR). The MCIP was calculated via discounted cash flow analysis, assuming a 10% IRR (Gong et al., 2025; Valasara et al., 2025b). The main economic assumptions and input parameters used in the TEA are summarized in Supplementary Material Tables S2–S6. Detailed discounted cash flow results for the four cases are provided in Tables S7–S10.

3.3.2. Break-even carbon price (BECP) and comparison with carbon allowance

Carbon allowance prices typically apply to the full CCS value chain, including CO_2 capture, conditioning, transport, and storage. However, as the system boundary of this study is limited to ship-based CO_2 transport and offshore injection, direct comparison between the MCIP and carbon allowance prices is not appropriate. Therefore, the BECP was introduced as an expanded economic indicator for carbon allowance comparison.

To enable a consistent comparison, a BECP was defined by adding the MCIP to fixed costs for CO_2 capture and liquefaction (Eq. (6)). These capture and liquefaction costs, adopted from representative literature, were assumed constant across all cases, as they are independent of the transport–injection configurations considered in this study. Thus, the BECP does not represent a fully modeled CCS chain, but rather combines the modeled transport–injection cost with literature-based upstream cost assumptions.

$$BECP = (CO_2\ capture\ and\ liquefaction\ Cost) + (MCIP) \tag{6}$$

3.3.3. Sensitivity analysis for MCIP

To enhance the robustness and interpretability of the eTEA results, a sensitivity analysis was conducted on key parameters. As emphasized in ISO/TS 14076, eTEA outcomes can vary significantly depending on input data and underlying assumptions, making sensitivity analysis a critical component for reliable interpretation. Transportation-related factors included transport distance, annual transport capacity, ship size, and fuel type. The analysis compared results under two conditions: (1) fixed ship size with varying transport distances, and (2) fixed annual injection volume with varying transport distances. Additionally, variations in fuel type were examined to assess trends when LCA outcomes were incorporated. The transport-distance sensitivity analysis was conducted by varying the route length while keeping the main ship operation assumptions consistent with the design basis summarized in Table 1. This approach was used to isolate the first-order effect of transport distance on the MCIP. Detailed ship design constraints, including fuel tank capacity, increased fuel weight, and the resulting reduction in effective CO_2 payload, were not explicitly modeled in this screening-level analysis and should be addressed in future detailed ship design and logistics studies.

3.3.4. Uncertainty analysis for eTEA

Uncertainty analysis is essential for quantifying the effects of variability in data and assumptions and for determining the confidence interval of the results (ISO, 2025). In this study, uncertainties were considered for capital costs for each equipment area and operating costs, including LNG fuel price, electricity cost, MEA cost, and carbon allowance price. Monte Carlo simulations were performed with assumed variations of 10%, 30%, and 50% to derive the distribution of the MCIP. Accordingly, both area-specific capital cost variability and market-related operating-cost variability, including fuel price and carbon credit valuation, were considered in the uncertainty analysis.

Each uncertain parameter was independently sampled using a triangular distribution. For each parameter, the baseline value was used as the mode, while the minimum and maximum values were defined according to the assumed variation range. Capital cost and operating cost uncertainties were evaluated separately to identify their respective effects on the MCIP distribution. A total of 10,000 iterations was conducted to ensure statistical robustness (Ahn et al., 2026b).

3.3.5. Subsidy-based sensitivity analysis

To evaluate the impact of cost reductions on the economic performance of the CO_2 transport and injection system, a subsidy-based sensitivity analysis was conducted. These reductions, while not intended to reflect specific policy instruments, were introduced to simulate improved project-level financial conditions.

Two types of cost reductions were considered. First, capital cost reductions were applied to the fixed capital investment (FCI) to assess the sensitivity of the MCIP to changes in upfront investment. Second, reductions in variable operating costs (VOC)—specifically fuel and electricity—were implemented to evaluate their relative influence on the MCIP. Each cost reduction was applied independently, with all other parameters held constant.

4. Results and discussion

4.1. LCA results

4.1.1. GWP comparison

Fig. 2 presents the gate-to-gate GWP results for each CO_2 transport–injection case. In Case 1 and Case 2, the GWP was identical at 21.87 kg CO_2/tCO_2 because both cases used the same LNG fuel consumption for maritime transport and the same electricity requirement for offshore injection. The GWP of these cases consisted mainly of LNG combustion emissions, 20.30 kg CO_2/tCO_2, with an additional 1.57 kg

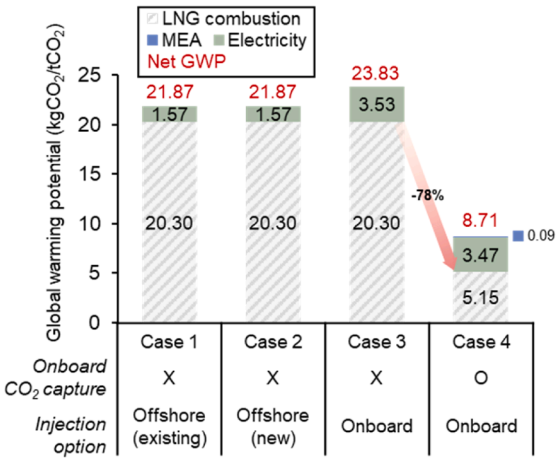


Fig. 2. GWP associated with each CO_2 transport and injection configuration within the gate-to-gate system boundary, expressed per 1 tCO_2 injected.

CO_2/tCO_2 from offshore injection electricity. In Case 3, the operation of onboard injection equipment required a higher electricity input, resulting in the highest GWP at 23.83 kg CO_2/tCO_2. This result indicates that electricity consumption during the injection stage can increase the environmental burden in addition to transportation-related emissions. The magnitude of this electricity-related burden may vary depending on the electricity supply source and regional grid carbon intensity. By contrast, Case 4 incorporated onboard CO_2 capture, leading to a substantial reduction in GWP to 8.71 kg CO_2/tCO_2. This represents a 63% decrease relative to Case 3, indicating that the MEA-based capture process effectively offsets emissions generated during the transport and injection stages.

4.1.2. Sensitivity analysis for GWP

Fig. 3 presents the results of the sensitivity analysis evaluating the effects of (a) transport distance and (b) fuel type on the GWP of CO_2 transport and injection.

In Fig. 3(a), Cases 1 and 2 recorded a GWP of 8.01 $kgCO_2/tCO_2$ at 2,000 km, which increased to 14.46 $kgCO_2/tCO_2$ at 4,000 km and 31.07 $kgCO_2/tCO_2$ at 6,000 km, before decreasing slightly to 27.34 $kgCO_2/tCO_2$ at 8,000 km. Case 3 exhibited a similar trend, with GWP rising from 7.52 $kgCO_2/tCO_2$ at 2,000 km to 15.08 $kgCO_2/tCO_2$ at 4,000 km, and to 32.85 $kgCO_2/tCO_2$ at 6,000 km, followed by a slight decrease to 30.31 $kgCO_2/tCO_2$ at 8,000 km. In contrast, Case 4 consistently exhibited the lowest GWP across all distances. Starting at 2.76 $kgCO_2/tCO_2$ at 2,000 km, it increased to 5.52 $kgCO_2/tCO_2$ at 4,000 km, 10.77 $kgCO_2/tCO_2$ at 6,000 km, and only slightly further to 11.06 $kgCO_2/tCO_2$ at 8,000 km. The increase in Cases 1 and 2 relative to the previous calculation reflects the additional electricity demand for offshore injection. These results indicate that, while longer transport distances increase fuel consumption and associated GWP emissions, a reduced number of trips due to higher transport capacities leads to nonlinear changes in GWP per tonne of injected CO_2. Notably, the onboard capture implemented in Case 4 effectively mitigated environmental impacts even at longer distances, suggesting clear advantages for large-scale, long-haul CCS shipping applications.

Fig. 3(b) compares the GWP associated with different fuel types across the cases. Fossil fuel-based options, including MGO, HFO, and LNG, exhibited higher GWPs than alternative fuels such as methanol and ammonia, primarily due to differences in CO_2 emission factors during combustion. MGO and HFO had the highest emission factors, at 3.21 and 3.11 $kgCO_2/kg$, respectively, followed by LNG at 2.75 $kgCO_2/kg$. Methanol was significantly lower at 1.38 $kgCO_2/kg$, while ammonia, producing no CO_2 during combustion, yielded 0 $kgCO_2/kg$. After including offshore injection electricity in Cases 1 and 2, the GWP ranged from 1.57 to 27.19 $kgCO_2/tCO_2$ across the fuel types and cases. These results confirm that the emission factor of the selected fuel is a critical determinant of overall environmental performance, while the fixed electricity demand for injection also contributes to the GWP. In particular, ammonia consistently resulted in the lowest GWP in all scenarios, underscoring its potential as the most environmentally favorable fuel option for ship-based CO_2 transport and injection. However, its application as a marine fuel remains in the early stages of commercialization, with several technical challenges yet to be addressed.

4.2. Energy and exergy analysis results

Fig. 4 presents the energy consumption and exergy distribution for the four CO_2 transport and injection configurations. The energy analysis (Fig. 4a) quantifies the total energy required to transport and inject 1 t of CO_2, while the exergy analysis (Fig. 4b) provides thermodynamic insight into how the supplied energy is utilized within the system.

As shown in Fig. 4(a), LNG fuel used for ship propulsion dominates the total energy consumption in all cases. In Cases 1 and 2, where ship transport and offshore injection are considered, the total energy consumption is identical at 367.0 MJ/tCO_2, consisting of 358.7 MJ/tCO_2 from LNG fuel and 8.3 MJ/tCO_2 from electricity used for offshore injection. In contrast, Cases 3 and 4 require additional electricity to operate onboard injection equipment and CO_2 capture systems, increasing the total energy consumption to 377.2 MJ/tCO_2 and 370.9

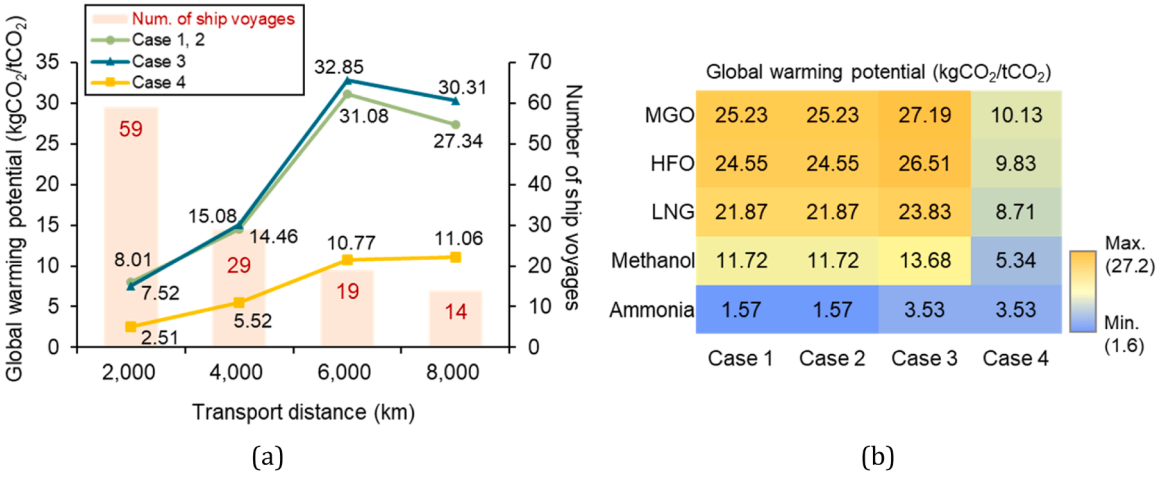


Fig. 3. Sensitivity analysis of GWP: (a) variation as a function of transport distance, and (b) comparison of fuel types across configurations.

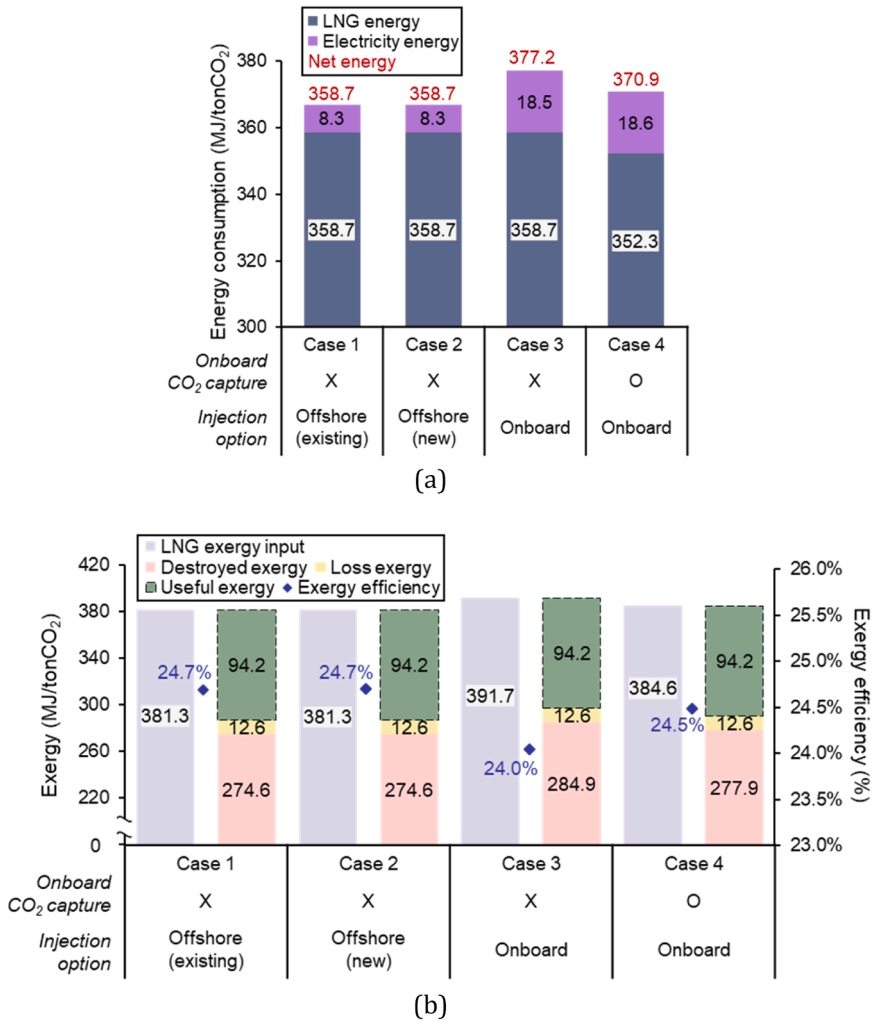


Fig. 4. Comparison of energy consumption and exergy distribution across the four CO_2 transport and injection configurations: (a) energy consumption and (b) exergy input, loss, destruction, useful exergy, and exergy efficiency.

MJ/tCO_2, respectively. Despite these differences, LNG fuel remains the dominant contributor to the total energy demand.

The exergy analysis provides further insight into the utilization of the supplied energy. As illustrated in Fig. 4(b), the total exergy input ranges from approximately 381 MJ/tCO_2 to 392 MJ/tCO_2 depending on the configuration. A large portion of the supplied exergy is destroyed due to thermodynamic irreversibilities, primarily associated with fuel combustion and propulsion processes. Across all cases, exergy destruction accounts for approximately 72–73% of the total exergy input. Exergy losses remain relatively small and nearly constant at about 12.6 MJ/tCO_2. In contrast, the useful exergy, defined as the physical exergy of the transported CO_2 stream, remains approximately 94.2 MJ/tCO_2 across all configurations. Consequently, the calculated exergy efficiency ranges from 24.0% to 24.7%. Cases 1 and 2 exhibit the highest efficiency at 24.7%, while Case 3 shows a slightly lower efficiency of 24.0% due to additional electricity consumption for onboard injection systems. In Case 4, the efficiency increases slightly to 24.5% as the reduced LNG fuel consumption partially offsets the additional energy required for onboard CO_2 capture.

These results are consistent with the GWP trends presented in Section 4.1. Since LNG combustion dominates the overall energy consumption, fuel use represents the primary source of CO_2 emissions. Consequently, configurations requiring additional energy input, such as Case 3, exhibit higher GWP values, whereas the integration of onboard CO_2 capture in Case 4 significantly reduces overall emissions despite the additional energy demand. Although Cases 1 and 2 include electricity use for offshore injection, ship propulsion fuel consumption remains the dominant contributor to both total energy demand and exergy destruction.

4.3. eTEA results

4.3.1. BECP

Fig. 5 presents the case-specific BECP for various CO_2 transport and injection scenarios, calculated by holding CO_2 capture and liquefaction costs constant while varying the downstream cost structure. Capture and liquefaction costs were treated as exogenous inputs, whereas all fuel consumption, operating costs, and capital-related expenditures were attributed solely to the CO_2 transport and injection subsystem. To address the substantial variability in capture costs due to differences in CO_2 concentration at the source, the analysis was performed separately for high-purity and low-purity CO_2 capture conditions. The projected carbon allowance prices shown in Fig. 5 were adopted from the IEA Global Energy and Climate Model Documentation 2024 (IEA, 2024), while the grey shaded area represents the observed 2023 Indonesia's NEK carbon allowance price range obtained from ICAP (ICAP, 2024).

Under the high-purity CO_2 capture assumption, the BECP was estimated at \$112.5/tCO_2 for Case 1, \$149.8/tCO_2 for Case 2, \$114.7/tCO_2 for Case 3, and \$118.0/tCO_2 for Case 4. Across all cases, the BECP was predominantly influenced by the fixed capture and liquefaction cost,

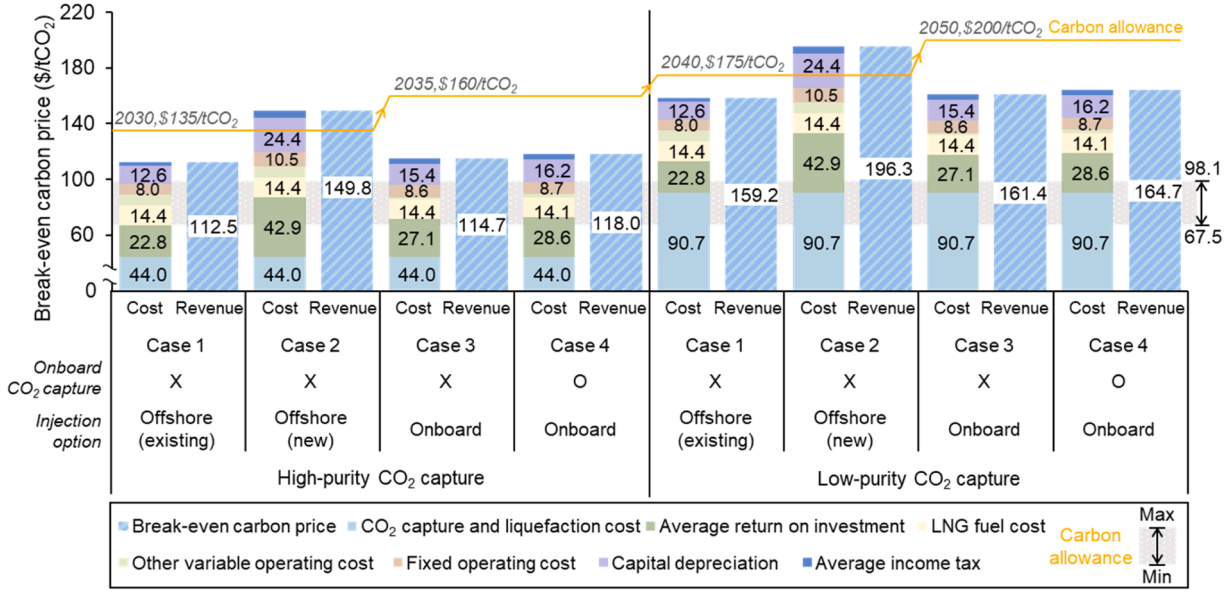


Fig. 5. Breakdown of costs and revenues comprising the BECP for high- and low-purity CO_2 capture scenarios, shown alongside carbon allowance price benchmarks for comparison.

along with capital-related expenditures for the transport–injection system, including capital depreciation and average return on investment. Case 2 exhibited the highest BECP, as the incorporation of offshore injection facilities led to capital-related costs comprising 44.9% of the total BECP. In contrast, Case 3 achieved a 23.4% reduction in BECP relative to Case 2, despite higher operating costs due to onboard injection, owing to significantly lower equipment investment compared to offshore injection infrastructure. In Case 4, the integration of onboard CO_2 capture introduced additional capital and operating costs; however, the associated CO_2 reduction offset these increases, resulting in BECP values comparable to those of Case 3.

Under the low-purity CO_2 capture condition, capture costs were approximately 106% higher than those in the high-purity case, resulting in a systematic increase in BECP across all scenarios. Despite this increase, the capital costs associated with the transport–injection subsystem remained the dominant component among non-capture-related expenditures. When compared with projected carbon allowance prices, all cases exhibited BECP values below the expected 2050 allowance level, indicating potential long-term economic feasibility. However, under near- to mid-term carbon allowance levels projected for 2030 and 2035, economic competitiveness varied across cases, underscoring the critical role of transport–injection system design in shaping intermediate-term cost performance. Accordingly, the following section presents sensitivity analyses with respect to MCIP, which captures the economic performance of the transport–injection subsystem, excluding fixed capture costs.

4.3.2. *Sensitivity analysis for MCIP*

Fig. 6 presents the sensitivity analysis of MCIP with respect to transport distance. Fig. 6(a) shows the results with fixed ship size, while Fig. 6(b) illustrates the case with a fixed annual CO_2 injection volume.

In Fig. 6(a), all cases exhibited a clear trend of decreasing MCIP with shorter transport distances. The distance range of 2,000–8,000 km was selected to cover short- to long-distance CO_2 shipping scenarios and to reflect long-distance transport cases considered in previous CO_2 shipping studies (DOE and NETL, 2025; Noh and Kang, 2025). This is attributed to the increased number of voyages enabled by shorter distances for a given ship size, which raises the annual CO_2 injection volume. Consequently, both fixed and fuel costs are distributed over a greater injected quantity, leading to a lower MCIP per tonne. For example, in Case 1, the MCIP decreased from \$88.5/tCO_2 at 8,000 km to \$22.0/tCO_2 at 2,000 km, representing a 75.1% reduction. Case 2 showed a decrease from

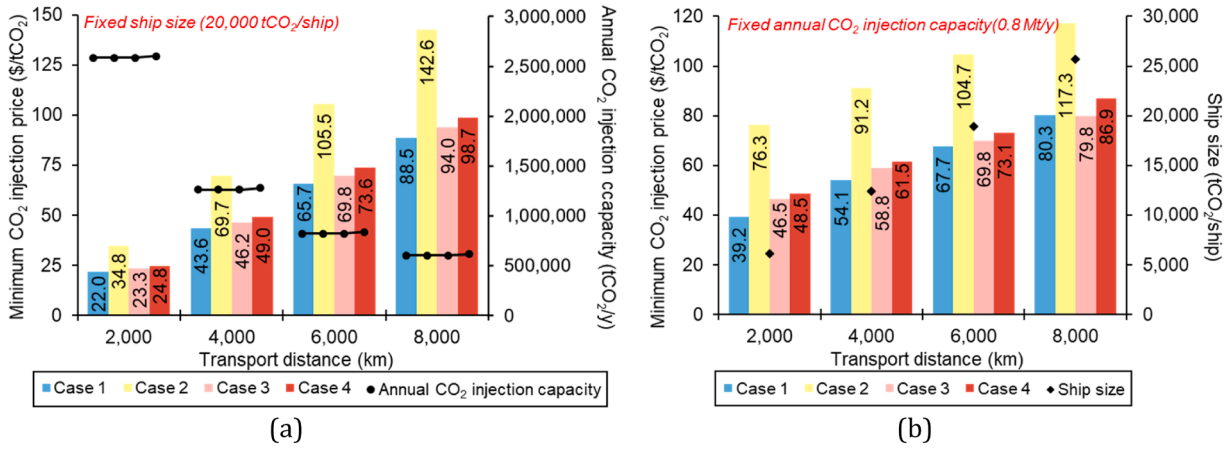


Fig. 6. Sensitivity analysis of the MCIP as a function of transport distance: (a) with fixed ship size and variable annual injection volume and (b) with fixed annual CO_2 injection capacity and variable ship size.

$142.6/tCO$_2$ to $34.8/tCO$_2$ (−75.6%), while Cases 3 and 4 exhibited reductions of 75.2% and 74.9%, respectively. These results indicate that, under a fixed ship size, shorter transport distances increase annual injection volumes and yield system-level economies of scale, thereby improving economic performance in short-range scenarios.

In contrast, Fig. 6(b) shows the results when the annual CO_2 injection volume is fixed. In this case, longer transport distances necessitate proportionally larger ship sizes to maintain constant annual delivery, thereby increasing both capital and fuel costs per voyage. As a result, the MCIP increases with distance in all cases. In Case 1, the MCIP rose from $39.2/tCO$_2$ at 2,000 km to $80.3/tCO$_2$ at 8,000 km, marking a 104.8% increase. Case 2 exhibited the smallest increase—from $76.3/tCO$_2$ to $117.3/tCO$_2$ (+53.7%)—due to its initially high MCIP, which diminished the relative effect of the cost increase. Cases 3 and 4 showed increases of 71.6% and 79.2%, respectively. Under fixed annual injection conditions, the cost advantage of shorter distances is eliminated, and MCIP increases almost linearly with transport distance.

Overall, transport distance was identified as the most influential factor affecting MCIP. As distance increases, the number of achievable voyages per year declines while fuel consumption rises, driving significant cost escalation. Annual injection volume—determined by the interplay between ship size and distance—serves to moderate these costs through economies of scale. In contrast, ship size influences economic performance indirectly via its effect on injection volume but has a comparatively limited impact relative to distance.

Fig. 7 presents the MCIP results for each case by fuel type, incorporating the LCA outcomes from Fig. 3(b) directly into the economic analysis.

In Case 1, HFO resulted in the lowest MCIP at $65.4/tCO$_2$, while ammonia yielded the highest at $88.9/tCO$_2$, approximately 36% higher than HFO. LNG, MGO, and methanol exhibited intermediate values, ranging from 5% to 12% above HFO. Case 2 produced the most unfavorable economic outcomes, with all fuels exceeding $100/tCO$_2$. Relative to Case 1, MCIP increased by 57% for HFO, 54% for LNG, 51% for MGO, 54% for methanol, and 42% for ammonia, indicating that the transport–injection conditions in Case 2 impose a substantial cost burden irrespective of fuel choice. Case 3 exhibited MCIP values comparable to those of Case 1, with differences within ±5% across all fuels, suggesting nearly equivalent economic implications. In Case 4, MCIP was slightly higher than in Case 1, with increases ranging from 5% to 9%, depending on the fuel. Across all scenarios, HFO consistently yielded the lowest costs, while ammonia was at least 30% more expensive than HFO, underscoring a clear cost disparity among fuel options.

When comparing across fuels, ammonia consistently recorded the highest MCIP in all cases, making it the least economically favorable option. In contrast, HFO maintained the lowest costs, while methanol exhibited MCIP values generally comparable to LNG. Although ammonia achieved the lowest GWP in the LCA, its low heating value and high fuel cost contributed to the highest MCIP.

Thus, while ammonia and methanol demonstrate environmental advantages in the LCA, HFO and methanol show stronger economic competitiveness in the eTEA results. This underscores that fuel selection plays a critical role in balancing environmental performance with economic feasibility in CO_2 transport–injection systems.

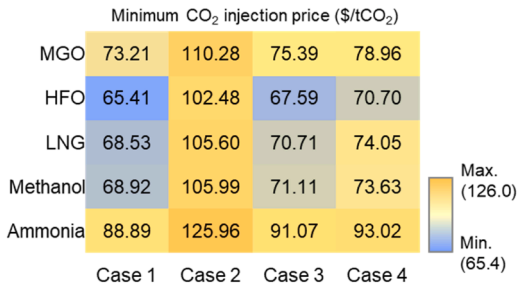


Fig. 7. Sensitivity analysis of MCIP by fuel type for each case.

Fig. 8 illustrates the influence of key economic parameters on the MCIP, including IRR, FCI, ship price, VOC, income tax rate, and BOG unit cost. These parameters were analyzed across ranges corresponding to both pessimistic and optimistic scenarios, as shown on the right-hand side of the graph.

Among the economic parameters, the IRR was the most sensitive factor across all cases. When the IRR decreased to 5% from the baseline scenario, the MCIP rose by 22.7% ($84.1/tCO$_2$) in Case 1, 27.9% ($135.0/tCO$_2$) in Case 2, 26.2% ($89.3/tCO$_2$) in Case 3, and 26.4% ($93.6/tCO$_2$) in Case 4. Conversely, when the IRR increased to 15%, the MCIP dropped by 20.3% ($54.6/tCO$_2$), 25.0% ($79.2/tCO$_2$), 23.5% ($54.1/tCO$_2$), and 23.7% ($56.5/tCO$_2$) in Cases 1 through 4, respectively. These results reflect how long-term cash flow discounting directly affects economic performance. The next most influential parameter was FCI, which exhibited symmetrical effects with ±10% deviations from the baseline. This resulted in MCIP variations of ±4.9% ($71.9/tCO$_2$, $65.2/tCO$_2$) in Case 1, ±6.2% ($112.1/tCO$_2$, $99.1/tCO$_2$) in Case 2, ±5.8% ($74.8/tCO$_2$, $66.6/tCO$_2$) in Case 3, and ±5.9% ($78.4/tCO$_2$, $69.7/tCO$_2$) in Case 4. The income tax rate had a greater impact on MCIP fluctuations than VOC in Cases 2–4, while Case 1—owing to its relatively smaller scale—showed only minor sensitivity to tax rate changes. Although taxation is a secondary factor compared with direct operating expenses, it can exert a stronger influence on MCIP in larger-scale systems.

In summary, the IRR and FCI were identified as the dominant factors influencing MCIP variation, whereas ship price, VOC, tax rate, and BOG unit cost had relatively limited impacts. These findings underscore that CO_2 transport–injection systems are inherently capital-intensive, with financial conditions and initial investment requirements serving as critical determinants of economic feasibility.

To contextualize the overall MCIP results presented in this section, the maritime route considered in this study was compared with reported pipeline- and ship-based CO_2 transport benchmarks. Previous studies indicate that the relative advantage of pipelines and ships depends strongly on transport distance, annual flow rate, infrastructure utilization, and project flexibility. IEAGHG (2020) reported that, for shore-to-offshore CO_2 transport with direct ship injection, ship-based transport becomes more cost-effective than pipeline transport above approximately 660 km for 1 $MtCO_2$/year and 990 km for 2 $MtCO_2$/year. Similarly, Nogueira et al. (2022) found that ships are generally more cost-effective for greater distances and smaller transport volumes, particularly above 400 km and below 10 $MtCO_2$/year, whereas pipelines are more suitable for larger volumes and shorter distances. The present study considers a long-distance cross-border route of approximately 6300 km and an annual transport volume of 0.79–0.81 $MtCO_2$/year, which falls well within the range where ship-based transport is commonly reported to be advantageous. Because published benchmarks differ in system boundary and cost assumptions, they are used here to contextualize the distance and volume conditions rather than to provide a direct cost-by-cost comparison. Accordingly, the maritime route evaluated here should not be interpreted as universally preferable to pipeline transport, but as a viable option for long-distance, cross-border offshore CCS deployment where pipeline construction, permitting, offshore pipeline routing, and large upfront infrastructure investment may limit practical deployment.

4.3.3. *Uncertainty analysis for MCIP*

Fig. 9 presents the results of the uncertainty analysis, in which parameters related to capital and operating costs were simultaneously varied to evaluate their impact on MCIP. Fig. 9(a) reflects the influence of all capital cost components included in each case—such as ship cost, BOG reliquefaction system, offshore injection facility, onboard injection facility, and CO_2 capture unit—while Fig. 9(b) illustrates the effects of operating cost components, including LNG fuel, electricity, MEA, and carbon credits.

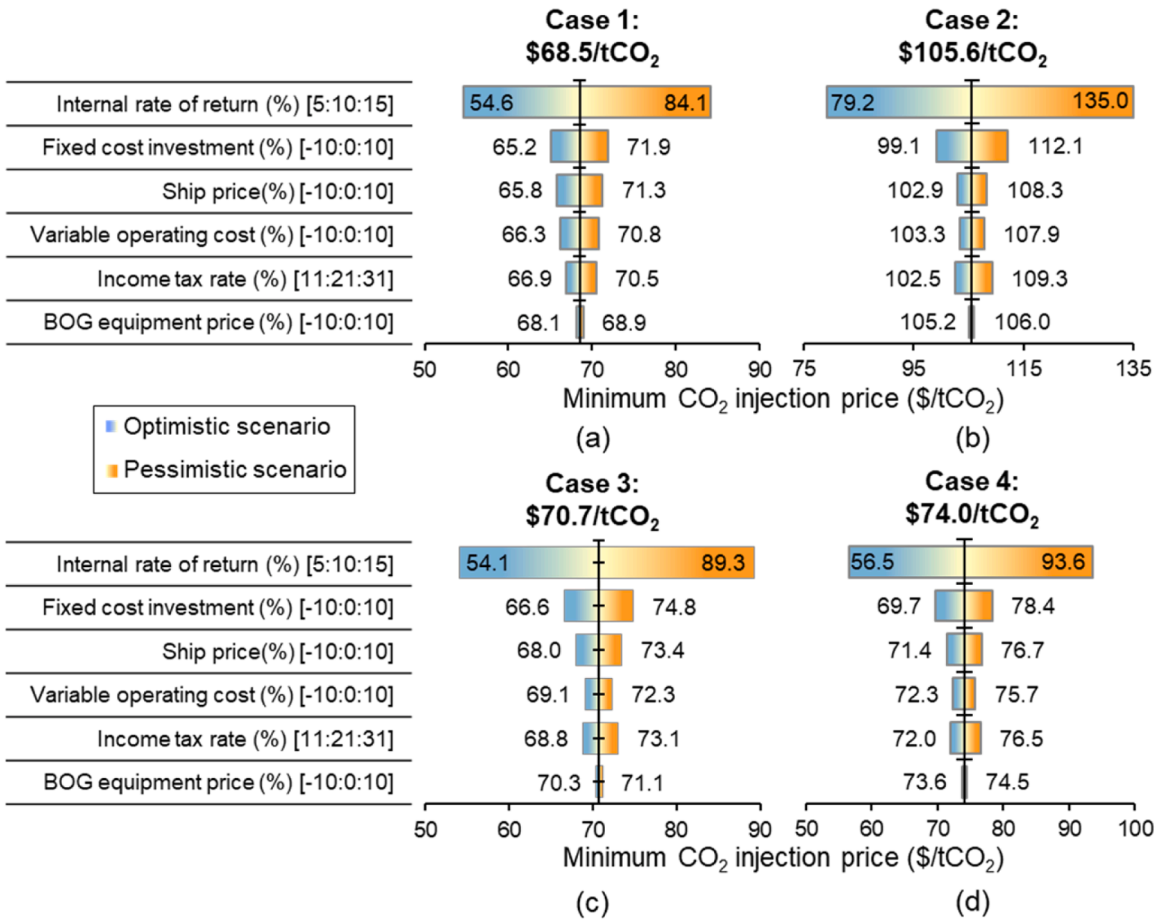


Fig. 8. Sensitivity analysis of the MCIP with respect to key economic parameters for (a) Case 1, (b) Case 2, (c) Case 3, and (d) Case 4.

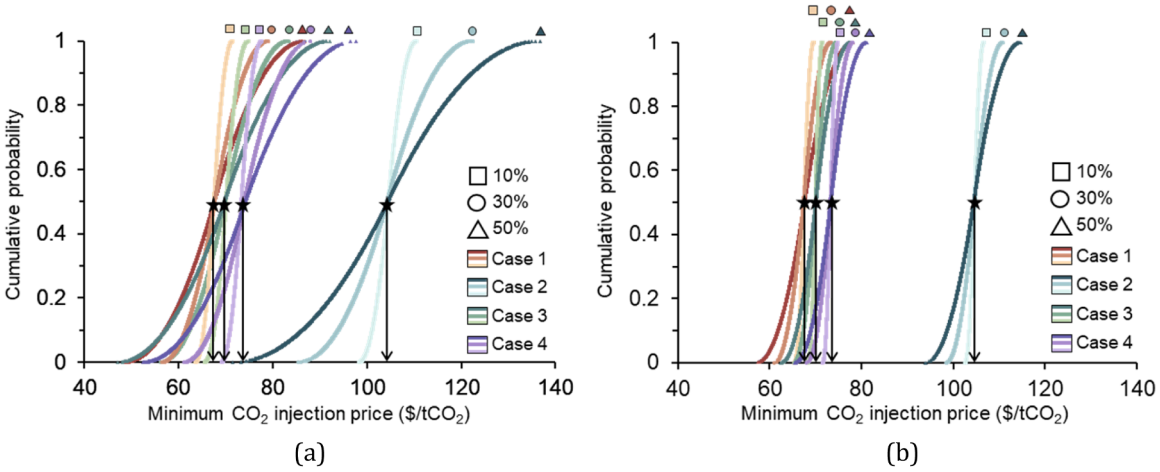


Fig. 9. Uncertainty analysis of the MCIP: (a) sensitivity to variations in capital costs; (b) sensitivity to variations in operating costs. Colors indicate Cases 1–4, and squares, circles, and triangles indicate the ±10%, ±30%, and ±50% uncertainty ranges, respectively.

In Fig. 9(a), variations in capital costs had a significant influence on MCIP across all cases. For Case 1, a ± 10% change in capital costs resulted in an MCIP range of \$63.6/tCO$_2$ (−5.8%) to \$71.5/tCO$_2$ (+5.8%). In Case 2, the range was \$98.3/tCO$_2$ (−6.0%) to \$110.5/tCO$_2$ (+5.7%); for Case 3, \$65.6/tCO$_2$ (−6.3%) to \$74.8/tCO$_2$ (+6.8%); and for Case 4, \$69.0/tCO$_2$ (−5.8%) to \$77.7/tCO$_2$ (+6.0%). These results suggest that moderate cost fluctuations (±10%) lead to relatively modest changes in MCIP. Under extreme uncertainty conditions (±50%), however, the MCIP range widened substantially. In Case 1, MCIP varied from \$48.5/tCO$_2$ (−28.1%) to \$86.8/tCO$_2$ (+28.5%); in Case 2, from \$71.6/tCO$_2$ (−31.6%) to \$136.7/tCO$_2$ (+30.6%); in Case 3, from \$47.1/tCO$_2$ (−32.8%) to \$92.1/tCO$_2$ (+31.6%); and in Case 4, from \$52.4/tCO$_2$ (−28.6%) to \$97.7/tCO$_2$ (+33.3%). These results underscore the high sensitivity of system economics to long-term capital investment, reinforcing the importance of minimizing initial capital expenditures to improve cost competitiveness.

Fig. 9(b) shows the impact of variations in operating costs and carbon credit values, which were comparatively less influential than capital

cost variations. Even under extreme uncertainty conditions (±50%), the MCIP variation remained within ±15.6% across all cases. Notably, Case 2 exhibited a relatively minor effect (from +9.6% to −10.3%) due to the overwhelming dominance of capital costs in its total cost structure.

In summary, the economic performance of the CO_2 transport and injection system is primarily determined by capital costs rather than operating costs. Among capital components, ship cost and offshore injection facility cost represent the largest shares, indicating that targeted improvements in these areas could substantially enhance overall price competitiveness.

4.4. Subsidy-based sensitivity analysis results

Fig. 10 illustrates the variations in MCIP and BECP for each case under different capital subsidy levels. To isolate the economic impact of the transport and injection infrastructure, the capture and liquefaction cost was held constant at \$67.3/tCO_2, representing the average of high- and low-purity CO_2 capture costs. The resulting BECP values were then compared with projected carbon allowance prices to evaluate the economic feasibility of each configuration.

Across all cases, MCIP decreased almost linearly with increasing capital subsidy ratios. Case 2, which includes an offshore injection facility, exhibited the highest sensitivity to capital subsidies due to its greater capital intensity. Its MCIP declined from \$105.6/tCO_2 at 0% subsidy to \$29.0/tCO_2 at 100% subsidy, corresponding to the largest reduction (72.5%). The MCIP at 100% subsidy represents a lower bound based solely on operating costs, as capital recovery is entirely excluded. In contrast, Cases 1, 3, and 4—characterized by lower capital requirements—showed smaller but still substantial reductions ranging from 57.7% to 68.6%, while maintaining lower absolute MCIP values across the entire subsidy range.

In terms of BECP, the cost-reduction effect of capital subsidies was more limited than that observed for MCIP. Although BECP decreased as the subsidy ratio increased, it followed an asymptotic trend, converging toward the fixed capture and liquefaction cost floor of \$67.3/tCO_2. This behavior highlights a structural limitation: reductions in transport and injection capital costs alone cannot reduce the total CCS cost below the threshold imposed by capture-related expenses.

Comparison with projected carbon allowance prices further clarifies the implications of capital subsidies. As indicated by the diamond markers in Fig. 10, the minimum subsidy required to achieve the 2030 projected carbon allowance price of \$135/tCO_2 varies significantly among configurations. Cases 1, 3, and 4 reach this threshold with relatively modest subsidies of approximately 20%, whereas Case 2 requires a considerably higher subsidy of around 60%. These results quantitatively distinguish the substantial effectiveness of capital subsidies in reducing transport and injection costs from their more limited influence on the overall CCS cost structure.

Fig. 11 illustrates the response of MCIP to variations in capital cost and VOC subsidy ratios for each transport–injection case. The x-axis represents the capital cost subsidy ratio (0–100%), while the y-axis denotes the VOC subsidy ratio (0–100%). Arrows in the figure indicate the relative change in MCIP resulting from a 50% reduction in each cost component.

Across all cases, a 50% reduction in capital costs led to substantial decreases in MCIP. In Case 1, the MCIP decreased by approximately 29%, while larger reductions of about 36%, 34%, and 34% were observed for Cases 2, 3, and 4, respectively. The greatest sensitivity occurred in Case 2, the most capital-intensive configuration, due to its inclusion of offshore injection facilities.

Reductions in VOC also decreased MCIP, although the effect was more moderate than that of capital cost reductions. A 50% reduction in operating costs led to MCIP decreases of approximately 17% in Case 1, 11% in Case 2, 12% in Case 3, and 11% in Case 4. These results suggest that while operating cost reductions contribute to improved economic performance, their impact is less significant than capital cost reductions.

Overall, the sensitivity analysis confirms that MCIP is significantly more responsive to changes in capital costs than to reductions in VOC. This trend becomes more pronounced with increasing capital intensity, emphasizing that upfront capital investment is the dominant factor influencing the economic viability of CO_2 transport and injection systems.

5. Conclusions

This study conducted an LCA of a ship-based CO_2 transport and injection system and integrated the results into the eTEA framework, in accordance with ISO/TS 14076. This represents the first application of the combined LCA–eTEA approach to ship-based CO_2 transport and offshore injection configurations, enabling a comparative evaluation of operational environmental performance and project-level economic performance within the defined system boundary. Comparison across the four cases demonstrated that the injection method and the presence of onboard capture were decisive factors influencing system performance. Case 4 achieved a GWP of 8.7 kgCO_2/tCO_2, corresponding to a 63% reduction relative to Case 3, attributable to onboard CO_2 capture. This underscores the strong environmental effectiveness of onboard capture in mitigating combustion-related emissions.

From an economic perspective, the analysis showed that cost performance is primarily governed by the capital structure of the

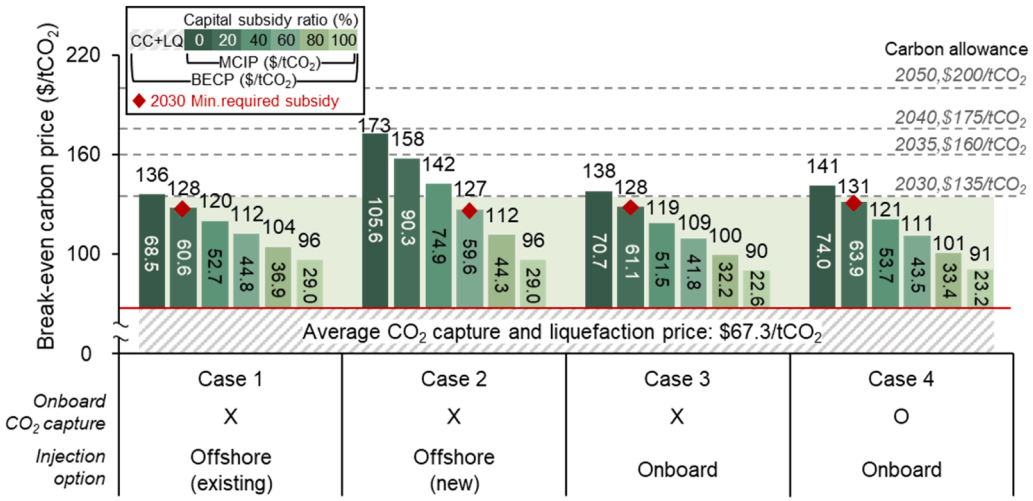


Fig. 10. Sensitivity analysis of MCIP and BECP as a function of capital subsidy levels under fixed CO_2 capture and liquefaction cost assumptions.

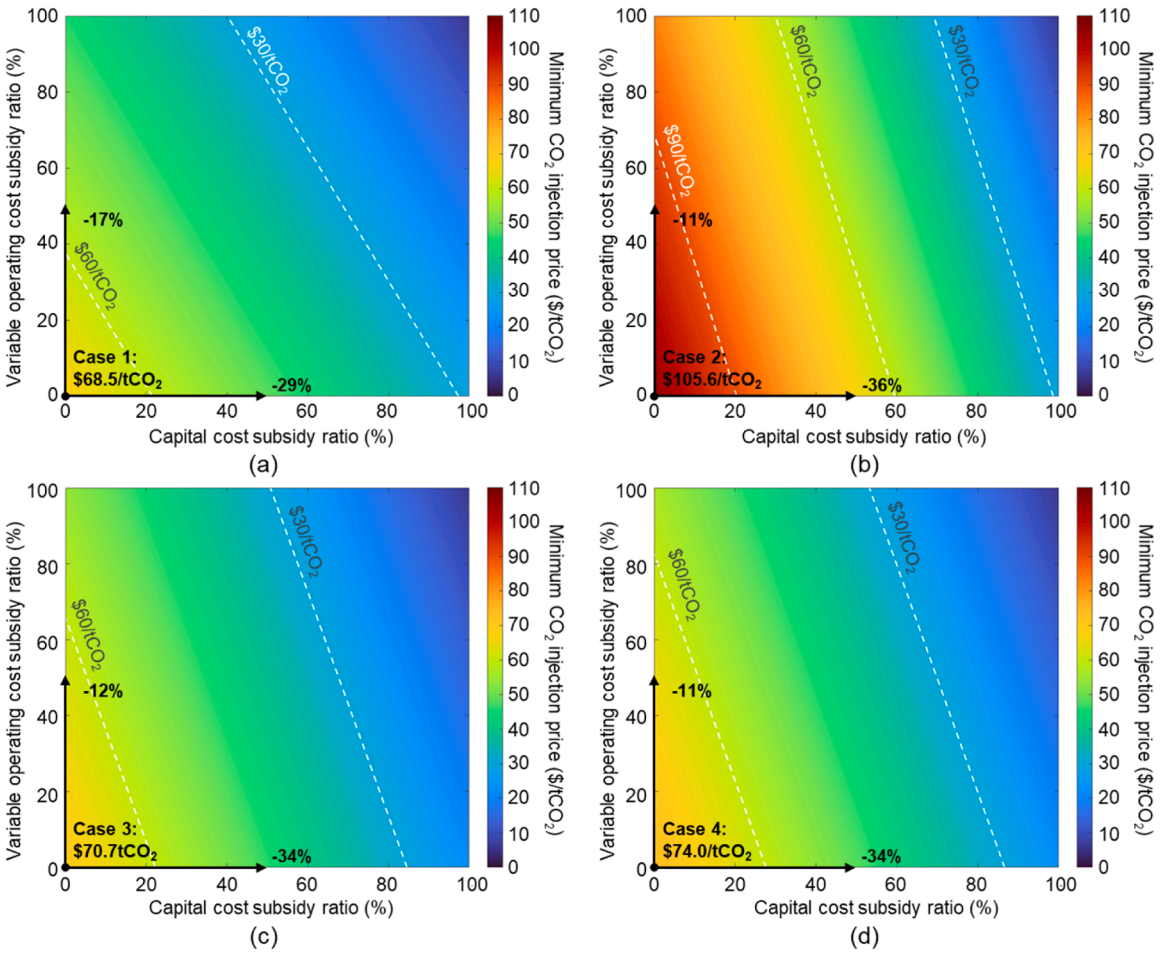


Fig. 11. Sensitivity analysis of the MCIP in response to capital cost and VOC subsidy ratios for (a) Case 1, (b) Case 2, (c) Case 3, and (d) Case 4.

transport–injection infrastructure. The MCIP was significantly more sensitive to capital cost variations than to changes in operating costs. Sensitivity analyses revealed that a 50% reduction in capital costs led to MCIP decreases of approximately 29–36% across the evaluated configurations, whereas an equivalent reduction in VOC resulted in more moderate decreases of 11–17%. Capital-intensive configurations, particularly those involving offshore injection facilities, exhibited the largest absolute cost reductions under capital cost support, indicating a strong dependence on upfront investment. These findings highlight that economic improvements in ship-based CO_2 transport and injection systems are most effectively achieved through optimization of capital-intensive components rather than marginal reductions in operating costs. The energy and exergy analysis results indicate that onboard injection can be implemented with only a small amount of additional electrical energy compared with the existing ship propulsion energy.

In conclusion, this study quantitatively demonstrated the environmental and economic performance of offshore CO_2 transport and injection systems by integrating LCA-derived indicators into the eTEA framework. The key implications are twofold: (i) onboard capture technologies play a central role in enhancing environmental outcomes, and (ii) economic viability is strongly influenced by capital-intensive infrastructure components. The findings also challenge the common perception that onboard capture provides limited benefits relative to its cost, showing that its emissions reduction potential can be substantial, particularly when deployed at scale. Taken together, these results highlight the necessity of the integrated consideration of system configuration and cost structure in the development of effective and economically viable ship-based CCS systems. Future work should extend the assessment to a full-chain CCS boundary by incorporating upstream CO_2 conditioning and compression, solvent degradation and end-of-life treatment, long-term storage monitoring, and detailed ship design constraints. The proposed assessment framework offers practical guidance for the design and evaluation of ship-based CO_2 transport and offshore injection projects under realistic operating conditions.

CRediT authorship contribution statement

Danbee Park: Writing – original draft, Visualization, Data curation. **Sang-Mok Han:** Writing – review & editing, Methodology, Conceptualization. **Namsub Woo:** Writing – review & editing, Methodology, Conceptualization. **Joonghun Kim:** Writing – review & editing, Methodology, Conceptualization. **Sungkyu Kim:** Writing – review & editing, Methodology, Conceptualization. **Jinwoo Park:** Writing – review & editing, Validation, Supervision, Project administration. **Wangyun Won:** Supervision, Project administration, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.psep.2026.109188.

References

Ahmed, S., Li, T., Li, S.Y., Chen, R., 2023. Comparative life cycle impact assessment of offshore support vessels powered by alternative fuels for sustainable offshore wind operations using machine learning. J. Ocean. Eng. Sci.

Ahn, B., Kwon, T., Ro, I., Won, W., 2025a. Valorization of plastic waste into drop-in fuels: Economic and environmental perspectives. Chem. Eng. J. 526, 171143.

Ahn, B., Jeong, S., Jin, G., Jeon, S., Na, J.-G., Won, W., 2025b. Production of ammonium formate using byproduct waste gas: economic and environmental perspective. ACS Sustain. Chem. & Eng. 13 (43), 18858–18871.

Ahn, B., Lee, D.-H., Lee, S., Chun, D., Chu, Y.-H., Park, Y.C., Won, W., 2026a. Integration of dry-sorbent CO2 capture process with natural gas combined cycle: Economic and environmental analyses. Energy Convers. Manag. 357, 121449.

Al Ghafri, S.Z., Perez, F., Park, K.H., Gallagher, L., Warr, L., Stroda, A., May, E.F., 2021. Advanced boil-off gas studies for liquefied natural gas. Appl. Therm. Eng. 189, 116735.

Al-Breiki, M., Bicer, Y., 2020. Comparative cost assessment of sustainable energy carriers produced from natural gas accounting for boil-off gas and social cost of carbon. Energy Rep. 6, 1897–1909.

Awoyomi, A., Patchigolla, K., Anthony, E.J., 2019. Process and economic evaluation of an onboard capture system for LNG-fueled CO2 carriers. Ind. & Eng. Chem. Res. 59 (15), 6951–6960.

Baek, Y., Gong, C., Yun, S., Lee, J.I., Won, W., 2026. Unveiling environmental benefits of carbon capture and utilization pathways: attributional and consequential life cycle assessments. Chem. Eng. J. 535, 175618.

Broker, Maersk, 2022. Marit. Transp. CO_2 Cost. Anal. 22500 M. ^{3} 7500 M. ^{3} Carr.

CATO, 2016. Transp. Unload CO_2 ship a Comp. Assess. (WP9 Final. Report.).

CLIMIT, 2022. Technology Qualification of Low-Pressure CO2 Ship Transportation. Retrieved from Climit website.

d'Amore, F., Natalucci, L., Romano, M.C., 2024. Optimisation of ship-based CO2 transport chains from Southern Europe to the North Sea. Carbon Capture Sci. & Technol. 10, 100172.

Deng, H., Roussanaly, S., Skaugen, G., 2019. Techno-economic analyses of CO2 liquefaction: Impact of product pressure and impurities. Int. J. Refrig. 103, 301–315.

DNV, 2021. CO2 offshore injection: Joint industry project.

DNV. (2025). Turning point for CCS is now, DNV report finds, with global capture and storage capacity expected to quadruple by 2030.

Element Energy, 2018. Shipping CO2–UK cost estimation study. Element Energy Limited, Cambridge.

Gür, T.M., 2022. Carbon dioxide emissions, capture, storage and utilization: Review of materials, processes and technologies. Progress. Energy Combust. Sci. 89, 100965.

Global CCS Institute, 2022. Glob. Status CCS 2022.

Gong, C., Na, H., Yun, S., Kim, Y.J., Won, W., 2025. Liquid hydrogen refueling stations as an alternative to gaseous hydrogen refueling stations: Process development and integrative analyses. Etransportation 23, 100386.

Gong, C., Yun, S., Woo, N., Han, S.-M., Won, W., 2026b. Unraveling the oil sands processing via partial upgrading: techno-economic and environmental perspectives. J. Clean. Prod. 553, 148003.

Gong, C., Ro, I., Yun, S., Kim, H., Woo, S., Won, W., 2026a. Green naphtha production via direct CO2 hydrogenation with renewable hydrogen: economic-environmental perspective. J. Environ. Manag. 397, 128384.

Han, J., Tan, Q., Ji, Q., Li, Y., Liu, Y., Wang, Y., 2023. Simulating the CCUS technology diffusion in thermal power plants: An agent-based evolutionary game model in complex networks. J. Clean. Prod. 421, 138515.

Holm, S.E., Winther, M.S., Straus, J., Roussanaly, S., 2025. How to best transport CO_2 to offshore storages? Carbon Capture Sci. & Technol. 15, 100416.

Hua, W., Sha, Y., Zhang, X., Cao, H., 2023. Research progress of carbon capture and storage (CCS) technology based on the shipping industry. Ocean. Eng. 281, 114929.

ICAP, 2024. Indonesian Economic Value of Carbon (NEK) Trading Scheme. International Carbon Action Partnership.

IEA. (2023). CO_2 emissions from international shipping in the net zero scenario, 2000–2030.

IEAGHG. (2020). The Status and Challenges of CO_2 Shipping Infrastructures. IEAGHG Technical Report 2020-10. IEA Greenhouse Gas R&D Programme, Cheltenham, UK.

IMO, 2023. 2023 IMO Strategy Reduct. GHG Emiss. Ships.

International Council on Clean Transportation (ICCT) (2025). Greenhouse gas emissions and air pollution from global shipping, 2016–2020.

International Energy Agency (IEA), 2024. Global Energy and Climate Model Documentation 2024. IEA, Paris.

International Organization for Standardization, 2006b. ISO 14044: environmental management — life cycle assessment — requirements and guidelines. International Organization for Standardization.

International Organization for Standardization, 2006a. ISO 14040: environmental management — life cycle assessment — principles and framework. International Organization for Standardization.

International Organization for Standardization, 2025. ISO/TS 14076: environmental management — environmental techno-economic assessment — principles, requirements and guidance. International Organization for Standardization.

IPCC, 2022. Climate Change 2022: Mitigation of Climate Change. Contribution of Working Group III to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change.

Jimenez, V.J., Kim, H., Munim, Z.H., 2022. A review of ship energy efficiency research and directions towards emission reduction in the maritime industry. J. Clean. Prod. 366, 132888.

Kim, T., Kim, Y., Park, J., 2026. Biogas-integrated liquid air energy storage system enabling negative carbon emissions and high energy efficiency. Energy Convers. Manag. 349, 120919.

Knoope, M.M.J., Ramirez, A., Faaij, A.P.C., 2013. A state-of-the-art review of techno-economic models predicting the costs of CO_2 pipeline transport. Int. J. Greenh. Gas. Control. 37, 214–233.

Lee, Y., Baek, K.H., Lee, S., Cha, K., Han, C., 2017. Design of boil-off CO2 re-liquefaction processes for a large-scale liquid CO2 transport ship. Int. J. Greenh. Gas. Control. 67, 93–102.

Leeson, D., Mac Dowell, N., Shah, N., Petit, C., Fennell, P.S., 2017. A Techno-economic analysis and systematic review of carbon capture and storage (CCS) applied to the iron and steel, cement, oil refining and pulp and paper industries, as well as other high purity sources. Int. J. Greenh. Gas. Control. 61, 71–84.

Luo, J., Xie, Y., Hou, M.Z., Xiong, Y., Wu, X., Lüddeke, C.T., Huang, L., 2023. Advances in subsea carbon dioxide utilization and storage. Energy Rev. 2 (1), 100016.

Ma, R., Zhao, Q., Wang, K., Cao, J., Yang, C., Hu, Z., Huang, L., 2025. Energy efficiency improvement technologies for ship in operation: A comprehensive review. Ocean. Eng. 331, 121258.

McKinsey & Company, 2025. CCS in Southeast Asia: An opportunity or distant reality.

Mitsubishi Heavy Industries, 2024. Joint Study to Establish Standard Specifications and Designs for LCO_2 Carriers.

Nogueira, T., Rochedo, P.R.R., Szklo, A., 2022. Evaluation of offshore CO2 transport alternatives in Brazil. Int. J. Greenh. Gas. Control. 116, 103575.

Noh, H., Kang, K., 2025. Techno-economic analysis of large-scale CO_2 ship transport with onboard boil-off gas reliquefaction. Int. J. Greenh. Gas. Control. 142, 104337.

Northern Lights, 2025. Northern Lights: Norway's first cross-border CO2 transport and storage project. TotalEnergies Retrieved from ⟨https://totalenergies.com/company/projects/carbon-capture-and-storage/northern-lights-norway⟩.

Oh, S., Kim, D., Kim, T., Park, J., 2025. Design of a hydrogen-water nexus by integrating autothermal reforming, electrolysis, and desalination. Energy Convers. Manag. 325, 119356.

Park, D., Lee, H., Won, W., 2024. Unveiling the environmental gains of biodegradable plastics in the waste treatment phase: a cradle-to-crave life cycle assessment. Chem. Eng. J. 487, 150540.

Park, D., Kim, H., Park, S., Lee, J., Kwak, D.H., Shin, K.S., Won, W., 2025a. Comprehensive life cycle assessment of powder-vs. Granule-form L-lysine production: Evaluating climate impact and sustainability. Chem. Eng. J., 162972

Park, J., Seo, K., Jeon, S., Lee, U., Won, W., 2026. Renewable olefin synthesis from flue gas: process development from a green perspective. J. CO2 Util. 108, 103459.

Park, M.-J., Yun, S., Jeong, D.-W., Won, W., 2025b. Upcycling municipal solid waste: Scenario-based technoeconomic evaluation and life cycle assessment. Energy 329, 136786.

Robalo-Cabrera, I., Alcayde, A., Filgueira-Vizoso, A., Castro-Santos, L., García-Diez, A.I., Manzano-Agugliaro, F., 2025. Shipping sector decarbonisation measures: A review. Sustain. Energy Technol. Assess. 82, 104549.

Roussanaly, S., Bureau-Cauchois, G., Husebye, J., 2014b. Costs benchmark of CO_2 transport technologies for a low-carbon future. Appl. Energy 116, 147–157.

Roussanaly, S., Brunsvold, A.L., Hognes, E.S., 2014a. Benchmarking of CO_2 transport technologies: Part II–Offshore pipeline and shipping to an offshore site. Int. J. Greenh. Gas. Control. 28, 283–299.

Rui, Z., Zeng, L., Dindoruk, B., 2025. Challenges in the large-scale deployment of CCUS. Engineering 44, 17–20.

S&P Global Commodity Insights (2024). Annual CO_2 shipping volumes seen at 100 mil mt/y by 2050: study.

Sou, W.S., Goh, T., Lee, X.N., Ng, S.H., Chai, K.H., 2022. Reducing the carbon intensity of international shipping–The impact of energy efficiency measures. Energy Policy 170, 113239.

Szargut, J., Morris, D.R., Steward, F.R., 1986. Exergy Anal. Therm. Chem. Metall. Process.

TGE Marine, 2023. CO_2 Carrier Brochure: Low- and Medium-Pressure Solutions.

U.S. Department of Energy (DOE) and National Energy Technology Laboratory (NETL), 2025. U.S.–Japan CO_2 Shipping Feasibility Study: Screening Assessment. U.S. Department of Energy, Washington, DC.

Vakili, S., Manias, P., Dbouk, W., Armstrong, L.M., Turnock, S., Teagle, D., 2023. IDRIC Proj. 50CO_2 Port. Pipeline (CO2P2P) CO2 Energy flows capture Marit. Transp. Offshore geostorage.

Vakili, S., Manias, P., Armstrong, L.M., Turnock, S., Teagle, D.A., 2025. Technical, economic, and environmental assessment of CO_2 ship transport in carbon capture and storage. J. Environ. Manag. 373, 123919.

Valasara, V., Ahn, B., Liu, J.J., Yu, B.Y., Han, S.M., Won, W., 2025b. Coproduction of Drop-in Biofuels and Biodegradable Plastic Monomers: Process Design and Integrative Analyses. ACS Sustain. Chem. & Eng. 13 (19), 6992–7004.
Valasara, V., Yun, S., Joo, J.C., Park, B.J., Won, W., 2025a. Production of renewable fuels and platform chemicals from carbon dioxide using microalgae: Economic and environmental perspective. J. Clean. Prod. 521, 146284.
Valasara, V., Ahn, B., Won, W., 2026b. Attributional and consequential life cycle assessments: a practice-oriented framework integrating system boundaries and machine learning. J. Chem. Eng. Jpn. 59 (1), 2662789.
Valasara, V., Yun, S., Joo, J.C., Lee, U., Park, B.J., Won, W., 2026a. Advances in plastic waste valorization: Economic and green perspectives. Renew. Sustain. Energy Rev. 237, 117031.
Visonà, M., Bezzo, F., d'Amore, F., 2024. Techno-economic analysis of onboard CO_2 capture for ultra-large container ships. Chem. Eng. J. 485, 149982.
Wang, Z., Porter, R.T., Zhang, F., Mahgerefteh, H., 2025. Techno-economic analysis of CO_2 conditioning and transport systems for offshore shipping. SSRN Electronic Journal. Elsevier BV.
World Economic Forum (WEF) (2024). Net-Zero Industry Tracker 2024: Shipping Sector.
Yang, Y., Seo, K., Kwon, J.S.I., Won, W., 2024. Process integration for the production of bioplastic monomer: techno-economic analysis and life-cycle assessment. ACS Sustain. Chem. & Eng. 12 (30), 11167–11180.
Yoo, B.Y., Choi, D.K., Kim, H.J., Moon, Y.S., Na, H.S., Lee, S.G., 2013. Development of CO2 terminal and CO2 carrier for future commercialized CCS market. Int. J. Greenh. Gas. Control. 12, 323–332.
Yun, S., Choe, B., Na, H., Ahn, B., Cho, H., Kim, Y.-J., Won, W., 2025. Improving the economics of sustainable aviation fuels: system-level analyses and perspective. ChemSusChem 18 (18), e70132.
Zahid, U., An, J., Lee, U., Choi, S.P., Han, C., 2014. Techno-economic assessment of CO_2 liquefaction for ship transportation. Greenhouse Gases Science Technology 4 (6), 734–749.
Zhang, Y., Li, X., Wang, J., et al., 2023. Research progress of carbon capture and storage (CCS) technology based on the shipping industry. Ocean. Eng. 284, 116474.
Zhao, T., Li, R., Zhang, Z., Song, C., 2025. Current status of onboard carbon capture and storage (OCCS) system: A survey of technical assessment. Carbon Capture Sci. Technol., 100402